Automatic System for Obstructive Sleep Apnea Events Detection Using Convolutional Neural Network

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***Abstract*— Obstructive Sleep Apnea (OSA) is characterized by repetitive episodes of airflow reduction (hypopnea) or cessation (apnea), which, as a prevalent sleep disorder, can cause people to stop breathing for 10 to 30 seconds at a time and lead to serious problems such as daytime fatigue, impaired memory, and depression. This work intends to explore automatic detection of OSA events with 1-second annotation based on blood oxygen saturation, oronasal airflow, and ribcage and abdomen movements. Deep Learning (DL) technology, specifically, Convolutional Neural Network (CNN), is employed as a feature detector to learn the characteristics of the high-order correlation among visible data and corresponding labels. A fully-connected layer in the last stage of the CNN is connected to the output layer and constructs the desired number of outputs for sleep apnea events classification. A leave-one-out cross-validation has been conducted on the PhysioNet Sleep Database provided by St. Vincents University Hospital and University College Dublin, and an average accuracy of 79.61% across normal, hypopnea, and apnea, classes is achieved.**

I. Introduction

Sleep is an important biological phenomenon. Its quality has a significant impact on the health of people who spend approximately one-third of their life on sleeping [1]. Obstructive Sleep Apnea (OSA), also referred to as Obstructive Sleep Apnea-Hypopnea (OSAH), is a potentially serious sleep disorder caused by complete or partial obstructions of the upper airway occurring when throat muscles intermittently relax and block the airway during sleep as shown in Fig. 1 [2], which is usually associated with a reduction in blood oxygen saturation [3]. This prevalent sleep disorder can cause people to stop breathing for 10 to 30 seconds at a time and lead to serious problems such as daytime fatigue, impaired memory, and depression. Patients with OSA are rarely aware of breathing difficulty, even when they are awake. It is often recognized by others via observing patients during episodes or suspected from its effects on their bodies [3].

Polysomnography (PSG) is the golden standard for diagnosing sleep apnea. Generally, an individual is monitored overnight with a signal acquisition device that collects electrical signals from a variety of sensors and electrodes attached to his/her body, such as oxygen saturation, electroencephalography and electrocardiography, which provide information about sleep state and respiratory details, as well as heart rate, body position and muscle contraction, etc. [4], [5]. Manually analyzing complex and large volume of PSG data from long recordings (usually 6-8 hours per night per subject), however, is a subjective process and a time-consuming task for professional sleep technologists.

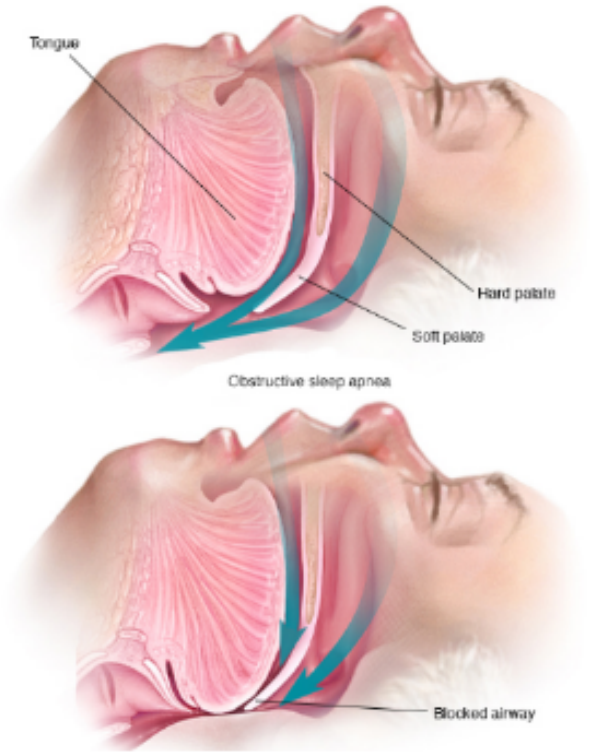


Fig. 1. OSA occurs when the throat muscles intermittently relax and block the airway during sleep [2].

With the rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) technologies, intelligent systems for automatic sleep analysis have attracted increasing attentions from both academia and industry. Based on various physiological signals, e.g. Electroencephalogram (EEG), electrocardiogram (ECG), blood oxygen saturation (SpO2),

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thoracic and abdominal signals, etc., OSA events detection systems have been developed based on various machine learning methods, such as Support Vector Machine (SVM) [6], [7], ensemble classifiers [8], Naive Bayes [9], k-Nearest Neighbor classification (KNN) [7], quadratic discriminant analysis [7], multivariate logistic regression [10], multiple logistic discrimination [11], Long Short Term Memory (LSTM) [12], [13], Deep Belief Network (DBN) [14].

PSG data are collected from various subjects with physiological characteristics, which makes data characteristics complex and leads to variant problems. When traditional shallow learning algorithms are employed, the success of event detection is heavily dependent on effective feature engineering. With the massively growing size of accessible data and the rapid development of computational power, Deep Learning (DL) has been successfully applied in many areas, e.g. natural language processing, imaging processing, speech recognition, biomedical signal processing, etc., and achieved significant improvement in comparison with other state of-the-art machine learning methods. Deep learning networks are able to identify structures in the lower-level data representations and provide organized features input for learning at the higher levels of data abstraction. This work intends to explore automatic detection of OSA events based on 1-second annotation by using DL technology. Convolutional Neural Network (CNN) is employed as a feature detector to learn the characteristics of the high-order correlation among multi-modal physiological signals and OSA event labels with a fully-connected layer in the last stage of the CNN linked to the output layer for sleep apnea classification.

The remainder of the paper is organized as follows. The CNN based learning model is presented in Section II, followed by experiment validation in Section III. Concluding remarks are given in Section IV.

II. Deep Learning based Events Classification Model

A. Physiological Signal

The multi-modal physiological signals used in our system include blood oxygen saturation, oronasal airflow, and ribcage and abdomen movements. Nasal airflow is the most common channel device used for obstructive sleep apnea detection. Patients receive less air during periods of apnea events, resulting in decreased oxygen delivery to the body denoted as oxygen desaturation. Ribcage and abdomen movements are used to measure patients mouth respiratory effort. A signal example is shown in Fig. 2.

The overflow artifacts of signals caused probably by bad calibration of the sensors are discovered by checking whether or not the signal under consideration reaches the limits of its dynamic range [15]. Correction of artifacts is carried out by linear interpolation. Each channel of signals is then normalized with zero-mean and unit-variance.

B. Data Annotation

Sleep databases are annotated by professional sleep technologists who label the onset time and duration of each

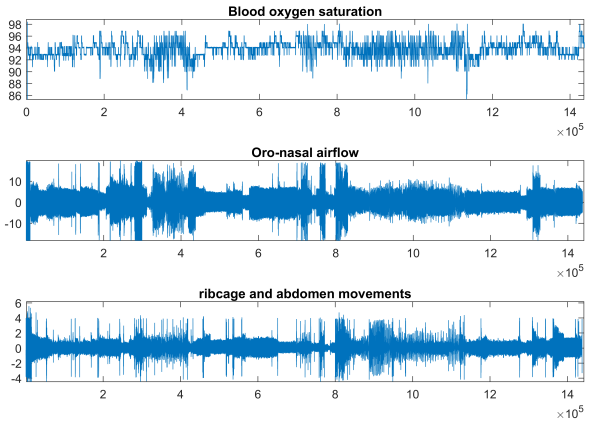


Fig. 2. Example of PSG signals in OSA event detection.

apnea and hypopnea event based on data recordings. In [16], [17], the reference annotation is defined on a 1-minute basis. In [16], two criteria are used to achieve the annotation of each minute. First, a single minute is considered as "apnea" if it contains 5 or more consecutive apnea and hypopnea events, otherwise this minute is labeled as "no apnea". Second, each minute is labeled as "apnea" when apnea is in progress at the beginning of the associated minute, otherwise this minute is label as "no apnea". In [17], the annotation was directly given to each of 1-minute segment by sleep technologists. Although 1-minute basis provides more time continuity information, compared to 1-second based annotation, it significantly reduces temporal resolution, and moreover, it reduces the number of data instances available for model training.

In our work, the annotation is given based on 1 second, that is, each second is labeled as one of 3 classes, i.e.

1) normal second;
2) hypopnea second (slow and shallow breathing with reduced oxygen supply to lungs due to partial obstruction to the airway path);
3) apnea second (temporary complete stoppage of breathing for 10 seconds or more).

C. Data Windowing

To capture sequential nature and temporal information, a temporal moving window is sliding throughout the time series signal of each channel. Let the length of the window be an odd value of Δ_w, the ith instance corresponding to i^{th} seconds is then built by centering on the ith seconds and containing $(\Delta_w - 1)/2$ previous seconds and $(\Delta_w - 1)/2$ subsequent seconds. Fig. 3 illustrates the data composition with a window of 5-second, in which each instance is started from 2 previous seconds, centered on the current second, and ended with 2 subsequent seconds. The number of signal samples in each instance can be expressed as

$$N_{sample} = \Delta_w \times Fs \times N_{channel}, \quad (1)$$

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where Fs and $N_{channel}$ are sampling frequency of signals and the number of channels used, respectively. If $\Delta_w = 5$, $Fs = 16$ Hz and $N_{channel} = 3$, we then have

$$N_{sample} = 5 \times 16 \times 3 = 240. \quad (2)$$

Each data instance built in this way is associated with one label (normal, hypopnea, or apnea) corresponding to the central second, which in the training set are fed into the classification model that will be presented in the following subsection. The labels of the instances in the testing set are then estimated based on the trained classification model.

D. Learning Model

PSG data used for OSA event detection are collected from various subjects having complex data characteristics and variant problems. When traditional shallow learning algorithms are employed, the classification performance is critically dependent on success of feature engineering. The advantage of DL networks is their abilities to identify structures in the lower-level data representations and extract organized features input for the learning at the higher levels of data abstraction.

Convolutional Neural Network is one of the most popular DL methods for discriminative deep architectures. A CNN is a feedforward neural network, in which each module consists of a convolutional layer and a pooling layer [18]. The pooling layer sub-samples the output of the convolutional layer and reduces the data rate from the layer below. The weights sharing in the convolutional layer and appropriately chosen pooling schemes endow the CNN with invariance properties (e.g. translation invariance), which have been effectively applied in computer vision and image recognition.

In this work, the CNN is first employed as a feature detector to learn the characteristics of the high-order correlation among the visible data and corresponding labels of OSA events. Using the CNN to automatically learn feature representation from raw data of physiological signals avoids the need of hand-crafted feature extraction and selection that are critically important in traditional shallow classification methods. In the CNN architecture used in our model, there are two stacks of convolutional and sub-sampling layers. The first convolutional layer has 6 feature maps and a kernel size of $5x5$, which is doubled in the second convolutional layer with 12 feature maps while using the same kernel size. The scale sizes in the 2 sub-sampling layers are both 2. After these, a fully-connected layer is linked to the output layer and constructs the desired number of outputs for sleep apnea classification as seen in regular neural networks. The output layer contains a single node for each target class, i.e. normal, hypopnea, and apnea classes.

III. Experiments

To evaluate the effectiveness of the developed method, the experiments was carried out on the PhysioNet Sleep Database that was provided by St. Vincents University Hospital and University College Dublin [19]. The dataset consists of 25 overnight PSG recordings from 21 male and 4 female subjects averagely aged 50 with suspected sleep disordered breathing. The 2 among 25 were abandoned due to their poor signal quality. The signals of blood oxygen saturation, oronasal airflow, and movements of ribcage and abdomen were originally sampled with a frequency of 8 Hz, and up-sampled to 16 Hz, each of which are normalized with zero mean unit variance.

The onset time and duration of respiratory events such as normal sleep, obstructive hypopnea, central hypopnea, mixed hypopnea, obstructive apnea, central apnea, mixed apnea, periodic breathing, and possible apnea or hypopnea, were annotated by the professional sleep technologists. Those seconds annotated as periodic breathing and possible apnea or hypopnea were abandoned and finally achieved a dataset with in total 24816 instances, each of which was labeled as hypopnea if they belonging to obstructive hypopnea, central hypopnea, or mixed hypopnea classes, or labeled as apnea if belonging to obstructive apnea, central apnea, or mixed apnea classes. The data proportion of normal, hypopnea and apnea is shown in Fig. 4, from which it can be seen that the data in different classes are quite imbalanced and 90% of instances belong to normal class.

Each instance stored in a vector with a length of 240 was reshaped into a matrix with a dimension of 16×15 and padded with zeros to get a matrix with dimension of 16×16 as input of the CNN. As discussed in Section II, there are one input layer, two stacks of convolutional and sub-sampling layer, one fully-connected layer, and one output layer in our network architecture. The first convolutional layer has 6 feature maps and a kernel size of $5x5$, and the second convolutional layer doubles the feature maps to 12 with the same kernel size, each follows a sub-sampling layers with a scale sizes of 2. A fully-connected layer is connected to the output layer that contains 3 nodes.

A leave-one-out cross-validation have been conducted, in which each time one acquisition was selected from all acquisition for the purpose of testing and the other 22 acquisitions were used for training the classification model. The average classification accuracies of normal, hypopnea, and apnea classes are 82.20%, 53.61% and 66.24%, respectively, with an average value of 79.61% across all classes.

IV. Conclusions

In this work, we intend to explore automatic detection of sleep apnea events based on blood oxygen saturation, oronasal airflow, and movements of ribcage and abdomen by using Deep Learning technology. Unlike other works with 1-minute basis, the developed system performs event detection based on 1-second annotation. Convolutional Neural Network, is employed as a feature detector to learn the characteristics of the high-order correlation among the visible data and corresponding labels and a classifier with a fully-connected layer connected to the output layer for sleep apnea classification. The leave-one-out cross-validation conducted on the PhysioNet Sleep Database achieved an average accuracy of 79.6% across normal, apnea, and hypopnea classes.

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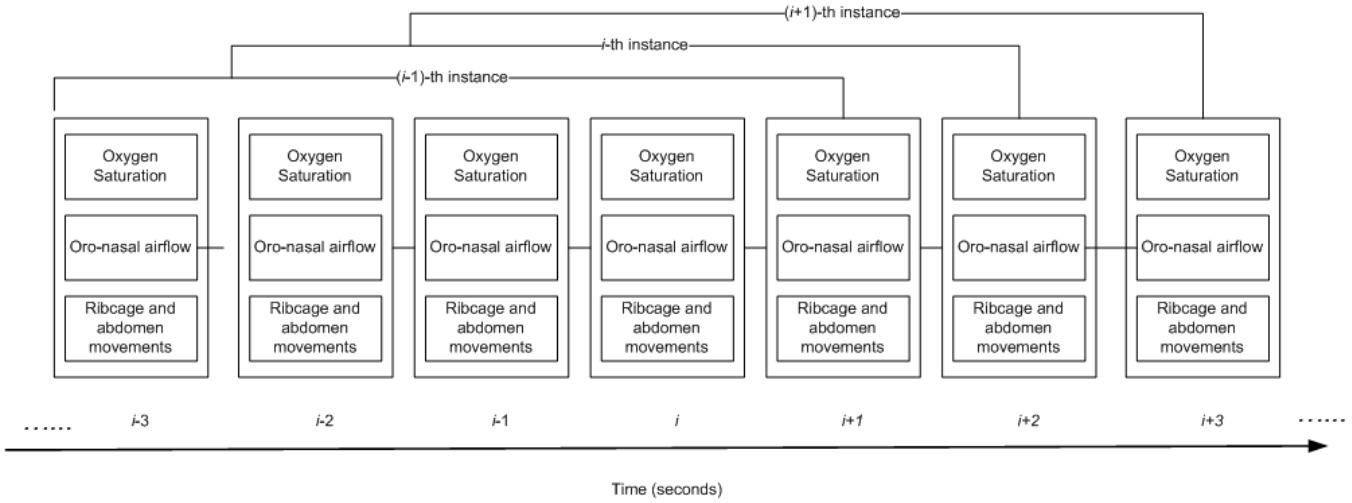


Fig. 3. Composition of each data instance.

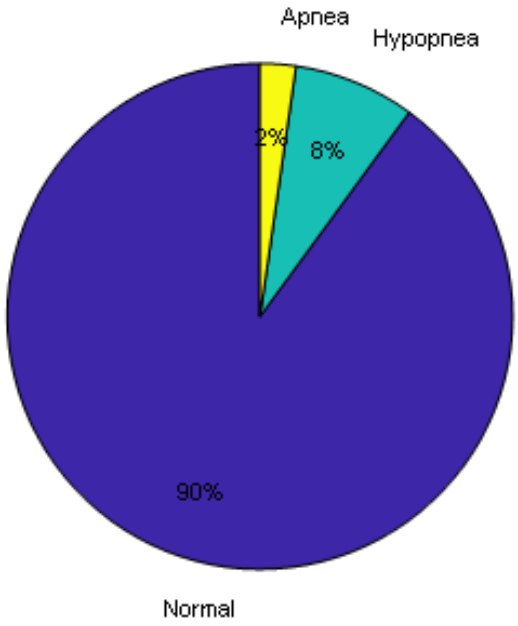


Fig. 4. Data proportion of sleep apnea events in the dataset.

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